

Full-production spectral discretization: integrated referee package

TECT verification-first repository · claim A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM

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1. Purpose and scope

This document is the self-contained referee package for the strongest P3 result currently justified by the evidence. The candidate closure scope is

$$\text{FROZEN-DISCRETIZATION-EVIDENCE-MATRIX.} \tag{1.1}$$

It covers the fixed three-torus with periods 16, 16, 16, the hash-pinned P1 three-component complex functional, the canonical P2 gradient flow with positive regularisers and $\eta_{\text{shell}} = 0$, the declared manufactured fields and Fourier grids, and one recorded CPU/CUDA precision environment.

The candidate does not claim that arbitrary historical N32/N64/N128 Sector-B solver trajectories carry a controlled continuum error. That stronger result requires an evaluated positive-time solution-ball constant and a dealiased finite-time evolution estimate, which remain under the open gate A3-FULL-DISCRETIZATION-CLOSURE.

1 Continuum and discrete objects

The field is $\Psi : \mathbb{T}^3 \rightarrow \mathbb{C}^3$, treated as six real components, with pairing

$$\langle u, v \rangle_{\mathbb{R}} = \text{Re} \int_{\mathbb{T}^3} u^\dagger v.$$

Let F be the P1 functional, $R = \nabla_{\mathbb{R}} F$, and let the P2 flow be $\partial_t \Psi = -R(\Psi)$. The linear part has symbol

$$r + Z|k|^2 + Y|k|^4 + M_{\text{fam}} + k_{\text{lock}}(I - P_0),$$

while the lower-order part contains the quartic/sextic potential and the positive-definite Class-II J - K quadratic form.

For the Fourier cube S_N , P_N is the real- L^2 projector. The exact Galerkin residual is $P_N R(\Psi_N)$. The canonical executable residual R_N^C is instead the real gradient of the periodic collocation energy. This package never identifies these two objects: their difference is the measured aliasing error.

2 Stage 1: spatial residual and real gradient

An analytic nonzero-density three-component trigonometric field activates all Class-II channels. Against N24 and N32 oversampled references, the N8/N12/N16 collocation-to-projected residual errors are

$$5.3301 \cdot 10^{-6}, \quad 4.2482 \cdot 10^{-10}, \quad 6.6679 \cdot 10^{-12}.$$

The observed rates are 23.28 and 14.44, reference uncertainty is at most $2.3351 \cdot 10^{-14}$, and the real energy-gradient identity is within $1.0211 \cdot 10^{-8}$. The frozen audit passes 13/13.

3 Stage 2: finite-time convergence

Forced manufactured evolution isolates RK4 time error on N8 and gives observed orders 4.016 and 4.008. The unforced N8/N12/N16 trajectories are compared with projected N20/N24 references at fixed final time; the relative error falls from $1.3063 \cdot 10^{-8}$ at N8 to $1.3118 \cdot 10^{-14}$ at N16. Time and space errors are recorded separately, and energy is nonincreasing on every declared grid. The audit passes 10/10.

4 Stage 3: Hessian and Ritz stability

At the homogeneous field, the declared twelve-real-dimensional Fourier block has a lowest isolated cluster with gap 0.1368657 and maximum Ritz-residual/gap about $1.01 \cdot 10^{-8}$. At the nonuniform Class-II stress field, the fixed-subspace Ritz errors are $1.5199 \cdot 10^{-3}$, $4.7888 \cdot 10^{-7}$, and $2.7136 \cdot 10^{-8}$ on N8/N12/N16. This is a block and fixed-subspace result, not a full-spectrum statement. The audit passes 13/13.

5 Stages 4 and 5: implementation and hardware independence checks

The portable implementation reconstructs the functional without calling the canonical energy/residual functions. Exact Fourier prolongation to M24/M32 then gives portable/canonical agreement and reference stability; the independent quadrature-proxy audit passes 11/11.

The recorded precision matrix covers CPU complex128/complex64 and CUDA complex128/complex64 at N8 on Torch 2.13.0+cu130 with one NVIDIA GeForce RTX 5070 Laptop GPU. CUDA complex128 has maximum energy/residual relative errors $5.7489 \cdot 10^{-16}$ and $1.7181 \cdot 10^{-15}$; CUDA complex64 gives $8.5245 \cdot 10^{-8}$ and $3.7843 \cdot 10^{-7}$. All six hardware assertions pass. No all-device or all-grid equivalence claim is made.

6 Stages 6 and 7: analytic solution-ball control

P2 gives a unique global H^2 solution and positive-time smoothing. For every H^2 initial radius R and $0 < \tau \leq T$, the fourth-order residual and the periodic aliasing estimate imply the qualitative bound

$$\sup_{\|u_0\|_{H^2} \leq R} \sup_{\tau \leq t \leq T} \|P_N R(u(t)) - R_N^C(P_N u(t))\|_{L^2} \leq C(R, \tau, T) N^{-2}. \quad (7.1)$$

The exponent/source audit passes 13/13.

The all-time H^2 input is explicit. A rigorous max-norm-shell Fourier embedding gives $\|u\|_\infty^2 \leq 0.1088517044 \|u\|_{H^2}^2$; P1 energy upper bounds and P2 coercivity then give

$$M_2(R)^2 = \frac{2\{E_+(R) + C_Y |\mathbb{T}^3|\}}{c_{H^2}}.$$

For $R = 0.5, 1, 2$, the enclosures are 20.16099, 20.35977, and 21.21616. The audit passes 13/13. These numbers are derived bounds, not trajectory fits.

7 Integrated reproduction map

The hash-bound verifier checks the immutable spatial, finite-time, Hessian/Ritz, independent-proxy, solution-ball-order, energy-ball, and recorded CUDA evidence at 79/79 PASS. The purpose-written PUBLISHED bundle, if confirmed, must include and execute the seven load-bearing audits and preserve every local manifest/result dependency. The principal repository command is

```
python codes/foundations/a3_full_production_verify.py --reuse-recorded-audits
```

and must print ASSERTS: 79/79 followed by A3-FULL-PRODUCTION-VERIFY-PASS.

8 Closure boundary and route to stronger tiers

The candidate T5 statement is only that the declared evidence matrix is closed, hash-pinned, independently implementable, and reproducible. Equation (7.1) remains qualitative because P2 uses unnamed norm-equivalence, local Moser, Holder, and endpoint-Duhamel constants. Therefore this candidate does not close full positive-time continuum error control.

To strengthen the result, one must expose these constants, compute $B_6(R, \tau, T)$ and $C(R, \tau, T)$, propagate them through a dealiased finite-time Galerkin stability inequality, and independently reproduce the resulting error budget. That is the route to a broader T5 scope and eventually a conditional T6 theorem; no T6/T7 claim is supported here.

9 Devil's-advocate review

1. **A frozen manufactured matrix is the same as a continuum theorem. UPHELD as false.** The candidate scope names the fields, grids, device, and unevaluated constant explicitly.
2. **Spectral differentiation removes nonlinear aliasing. UPHELD as false.** Pointwise nonlinear products alias, and R_N^C is kept distinct from $P_N R$ throughout.
3. **Fast N8/N12/N16 convergence guarantees all rough P2 data. UPHELD as false.** The analytic test field is smoother than a generic H^2 datum; only the qualitative positive-time bound is uniform over H^2 balls.
4. **One RTX 5070 run validates every accelerator. VALID WITH MITIGATION.** The exact recorded device and Torch version are part of the scope, and broader hardware equivalence is excluded.
5. **The Hessian block proves stability of a Sector-B candidate. UPHELD as false.** It certifies only the declared invariant block and fixed Ritz subspace at the tested fields.
6. **A bundle can be published before operator review. UPHELD as a governance violation.** This v2.0 package remains pending operator confirmation; no PUBLISHED bundle or T5 promotion is authorised by the document itself.

10 Falsifier

The candidate closure fails if a pinned source hash drifts, any of the 79 integrated assertions fails, an oversampling reference becomes unstable, the portable/canonical comparison exceeds its threshold, a declared precision row fails, or the bundle cannot reproduce the expected PASS outputs from its own copied dependencies. Such a failure returns the claim to T4 or lower; it is not absorbed by widening a tolerance after the fact.

Result footer

Result ID:	A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM
Precise statement:	Closed frozen discretization evidence matrix for the canonical full-production PDE and declared tests.
Scope:	T5 CANDIDATE@FROZEN-DISCRETIZATION-EVIDENCE-MATRIX; fixed torus, declared grids/fields, recorded device.
Dependencies:	A1 production functional; A2 T6 well-posedness.
Evidence grade:	ANALYTIC + EXECUTED + CONDITIONAL.
Reproduction command:	python codes/foundations/ a3_full_production_verify.py --reuse-recorded-audits

Expected output: ASSERTS: 79/79;
 A3-FULL-PRODUCTION-VERIFY-PASS.
Falsification gate: A3-FULL-DISCRETIZATION-CLOSURE remains open for the
 stronger quantitative continuum-error statement.
Tier before / after: T4 / T5 CANDIDATE (operator confirmation pending).
No-overclaim statement: Not a numerical $C(R, \tau, T)$, arbitrary-trajectory
 continuum certificate, full-spectrum theorem, T6/T7.
Next required action: Operator review/confirmation, then PUBLISHED bundle;
 separately derive B6/C and dealiased evolution.